ELECTRONS ON THE MOVE - SOME ELECTROCHEMISTRY

OBJECTIVES

- Gain experience with redox chemistry.
- Get comfortable in writing half reactions and using these reactions to calculate the reactions electrical potential, E°.

INTRODUCTION

The classical illustration of a reduction/oxidation (redox) reaction is the copper/zinc "couple" described by the two "half reactions"

$$Cu^{2+} + 2e^{-} \rightarrow Cu_{(s)} \qquad E^{\circ}_{red} = +0.337\,V$$

$$Zn^{2+} + 2e^{-} \rightarrow Zn_{(s)} \qquad E^{\circ}_{red} = -0.763\,V$$

written as reductions. V is voltage, or electrical potential: 1 volt is 1 J of energy per coulomb of electrical charge ($1\,V = 1\,J/C$) carried by the electrons in the reaction as described below. In the spontaneous direction, these half-reactions combine so that the copper ions are reduced (that is, written as shown, and is a reduction reaction) and the solid zinc becomes oxidized (written in the reverse direction, as an oxidation reaction) as

$$Zn_{(s)} \rightarrow Zn^{2+} + 2e^{-} \qquad E^{\circ}_{ox} = +0.763$$

When we add these two half-reactions together, we get the overall cell-reaction:

$$Zn_{(s)} + Cu^{2+} \rightarrow Zn^{2+} + Cu_{(s)} \qquad E^{\circ}_{cell} = +1.100\,V$$

This reaction is spontaneous as written in the forward direction, and generates an electrochemical voltage or electrical potential, of +1.100 V. The $Zn_{(s)}$ is in practice a piece of solid Zn metal called a Zn "electrode". This copper/zinc couple then has a zinc electrode that is gradually being corroded away and a copper electrode ($Cu_{(s)}$) that is growing with new solid Cu atoms as the reaction proceeds.

The reaction potentials above are the potentials for reactions at standard conditions (25℃ and 1 M concentrations). Standard potentials for many half-reactions are listed in tables (like those found on pages 10-11). These standard potentials can be used to predict the spontaneity (and thus feasibility) of all combinations of the half-reactions, providing a very useful tool for understanding a wide range of biochemical, environmental, geological and industrial reactions.

Voltage, or electrical potential (E), is not really a new concept because a volt of electrochemical potential is simply one Joule of free energy (ΔG) per coulomb of electrical charge moved from one chemical species to another. Each mole of electrons has (or carries, as we say) 96,485 coulombs; this charge is called the Faraday constant (F) which represents the electrical charge carried by one mole of electrons. Free energy is related to electrical potential through the following equation:

$$\Delta G = -n \times F \times E$$

where n is the number of moles of electrons written in a reaction (or half-reaction). When ΔG is negative (spontaneous reaction in the forward direction), the voltage is positive. For the copper/zinc redox couple with $[Cu^{2+}]$ = $[Zn^{2+}]$ = 1.0 M described above, $E^{\circ}_{cell} = +1.100\ V$ corresponds $\Delta G^{\circ} = -212300\ J = -212.3\ kJ$, consistent with a spontaneous forward reaction.

The actual E for a reaction will depend upon the concentrations for the various aqueous species in the reaction, namely, upon how the reaction quotient Q compares to the reaction's equilibrium constant K. The relationship is

$$E = \frac{RT}{nF} \ln\left(\frac{Q}{K}\right)$$

where R is $8.314\ \frac{J}{K \times mol}$, T is temperature in Kelvin, F is the Faraday constant, and the equilibrium constant K may be either K_c or K_p. Remember that a volt is $1\ \frac{J}{coloumb}$. When $Q = K$, i.e., reaction at equilibrium, the actual voltage $E = 0\ Volts$.

A little rearranging gives the Nernst equation

$$E_{actual} = E^{\circ}_{cell} - \frac{RT}{nF}\ lnK$$

where the standard cell potential is calculated by

$$E^{\circ}_{cell} = +\left(\frac{RT}{nF}\right)\ lnK$$

The equation above indicates that the equilibrium constant is proportional to the standard potential of the reaction. When K > 1 and $E^{\circ}_{cell} > 0$, the reaction is spontaneous and favors product formation. When K < 1 and $E^{\circ}_{cell} < 0$, the reaction is non-spontaneous and the reaction favors reactants formation. You can also think about this in terms of Le Chatelier's Principle, which states that when a system at equilibrium experiences a disruption, the system will minimize that change by shifting the equilibrium in the opposite direction. So, when you disrupt the concentration of a reactant of product away from standard conditions, you will see the equilibrium shift to minimize that change.

PROCEDURE

Some favorite chemical demonstrations involve reduction-oxidation reactions. This exercise provides you the opportunity to perform some of the demonstrations for your own enjoyment and to gain experience with some good redox chemistry. You should write balanced half reactions and full reactions for each chemical change you witness, and you need to fully describe all your observations.

Parts A - D of this laboratory can be done in any order.

A. Activity Series for Some Metals

1. Obtain a Petri dish and place it on a sheet of paper labeled with Cu, Fe, Sn, Zn, and Mg in a circular layout (see Figure 1).

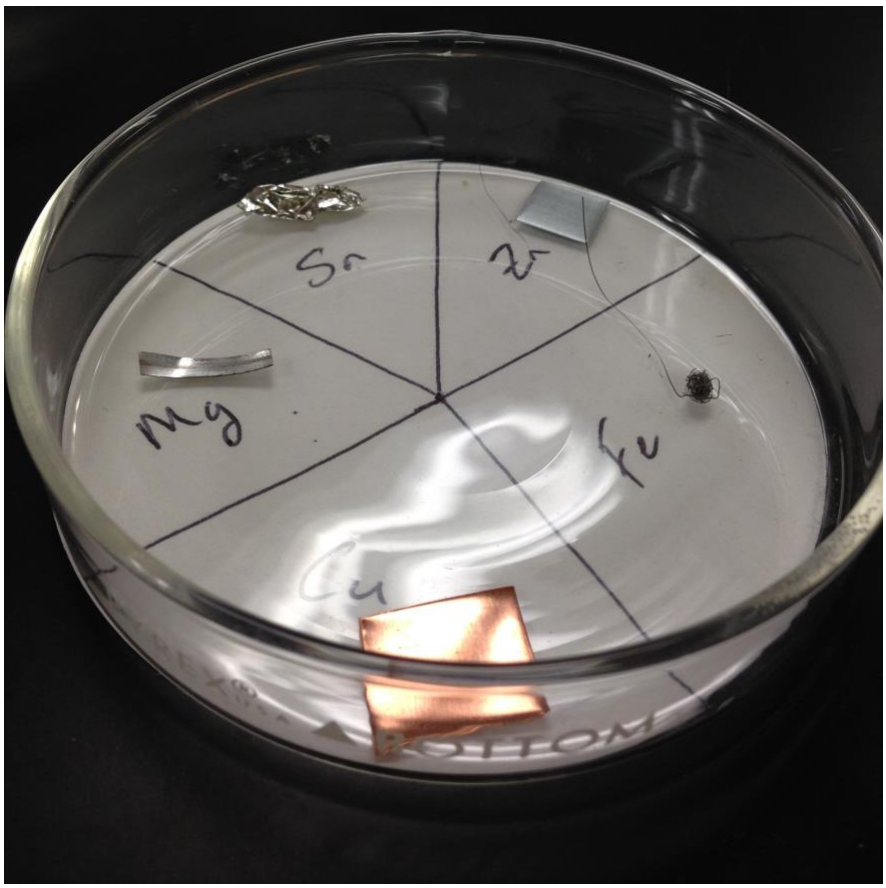


Figure 1: Arranged metal pieces in a Petri dish sitting on top of a sheet of paper with the elemental symbol of metals.

2. **IN THE HOOD**, pour and cover the bottom of the dish with 6 M HCl.
3. Place small pieces of Cu, Fe, Sn, Zn, and Mg in the dish near their respective label. If the metals are floating, poke down with a glass rod to start the reaction.
4. Note the extent of reaction of each metal with the HCl.

5. Write the reactions for what you observed and calculate E°_{cell} from the table of standard reduction potentials (Table 16.1) for each reaction.

Table 1. Half-reactions for metals reacting with HCl to generate overall cell-reaction and calculate E°.

Metal	**Half-reactions and Cell-Reaction**	$\boldsymbol{Total\ E^{\circ}_{cell}}$ **(V)**
Cu		
Fe		
Sn		
Zn		
Mg		

6. Correlate your findings with the relative positions of these metals on the reduction half reaction table found at the end of this manual.
7. Pour the acid solution and all metals into the waste container. Carefully rinse the Petri dish into the waste container using a squeeze bottle.
8. Wash the Petri dish with soap and water.

B. Cuperating an Aluminum Can

1. Insert a file into an aluminum soft drink or adult beverage can and scratch a circle around the circumference of the can's interior.

2. Fill the can with $1M\ CuCl_2$ solution, place the can in a 1.0 L beaker to catch any solution that leaks from the can as the reaction takes place.



Figure 2: Part B set up.

3. Let it stand for about five minutes.
4. Carefully pour the solution back into its original container (to be used again) and rinse the can.
5. Hold the top of the can in one hand and the bottom of the can in the other and pull the halves apart with a twisting motion.
6. Inspect the interior edges of the tear.
7. Write reactions for what you observe and calculate E°_{cell} from the table of standard reduction potentials (Table 16.1) for each reaction.

Table 2. Half-reactions for metals in a single-displacement reaction to generate overall cell-reaction and calculate E°.

Half-reactions	$\boldsymbol{E^{\circ}_{cell}}$ **(V)**
Cell-reaction	**Total** $\boldsymbol{E^{\circ}_{cell}}$ **(V)**

8. Put the used can in the box in the hood. Wash the 1 L beaker with soap and water.

C. Oxidation of Sodium Metal by Water

WARNING: DO NOT USE A LARGER PIECE OF SODIUM THAN CALLED FOR. THINK OF THE SIZE OF A PEA. DO NOT HANDLE SODIUM WITH YOUR BARE HANDS. USE TONGS OR A SCOOPULA. $NaOH$ IS CAUSTIC. WEAR GOGGLES AND GLOVES. ASK FOR A SAFETY SHIELD IF YOU FEEL NERVOUS ABOUT DOING THIS.

1. Half fill a 250 mL beaker with water. Add a few drops of phenolphthalein and observe the approximate pH of the solution after the reaction has occurred.
2. **IN THE HOOD**, nest a piece of filter paper in a weighing dish. Use this nest to transport a piece of freshly cut Na metal (**NO LARGER THAN 2mm x 2mm x 2mm**).
3. Immediately transfer the Na to the surface of the water in the beaker using tongs or scoopula. Observe the result.
4. **IN THE HOOD,** you may wish to repeat the reaction, this time having a burner and a burning wooden splint to ignite the gas being produced by the reaction of Na and water.
5. Repeat the process as many times as you wish. Don't get brave and try larger pieces. (There is no end to that temptation.)
6. Write the reaction for what you observe and calculate E°_{cell} from the table of standard reduction potentials (Table 16.1) for each reaction.

Table 3. Half-reactions for sodium reacting with water to generate overall cell-reaction and calculate E°.

Half-reaction	$\boldsymbol{E^{\circ}_{cell}}$ **(V)**
Cell-reaction	**Total** $\boldsymbol{E^{\circ}_{cell}}$ **(V)**

7. Pour the used water in the 250 mL beaker into the waste container. Wash the beaker with soap and water.

D. Measuring the Potential of a Redox Reaction

We can easily measure the potentials of redox reactions with Logger Pro and thus get a quantitative measure of the tendency for the reaction to proceed. Your results of Part A, show qualitatively what metals react most easily. On the basis of this information, we can predict just how spontaneous the reaction between any two of the metals will be. Here you will set up a simple redox couple in the form of an electrochemical cell using zinc and copper. Observing the voltage of the cell will measure the spontaneity of the reaction.

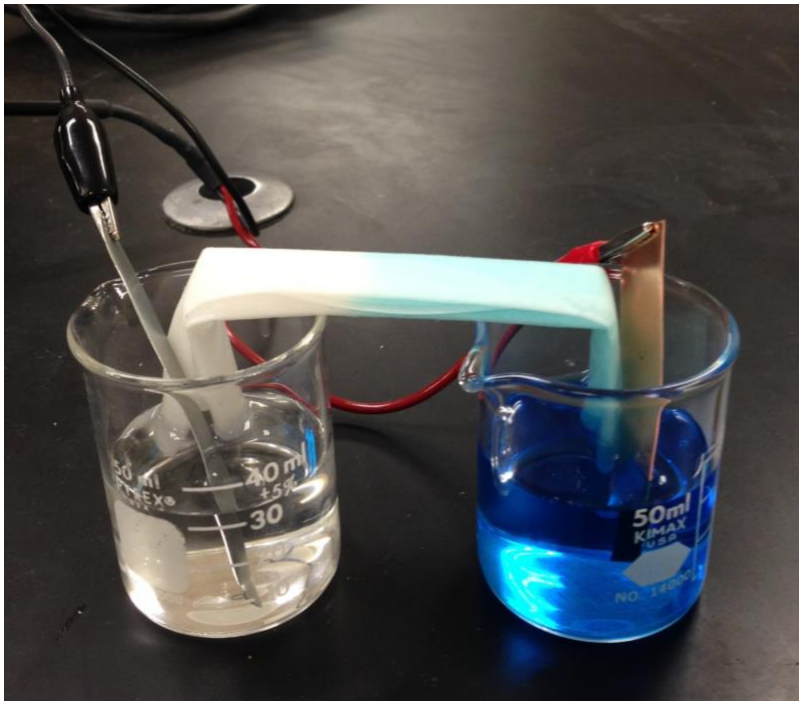


Figure 3: Electrochemical cell set up

1. Obtain two 50 mL beakers and a piece of 15 cm filter paper.
2. Pour about 35 mL of 1 M copper(II) nitrate solution in one and 35 mL of 1 M zinc(II) nitrate solution in the other.
3. Obtain strips of zinc metal and copper metal to use as electrodes. Note that these are different than the metals you used for Part A. They must be long enough to dip into the solution but keep the leads of the voltage probe out of the solution.
4. Prepare a salt bridge from the 15 cm filter paper: roll it up, bend it, and insert one end in the copper solution and the other end in the zinc solution. Carefully pour a minimal amount of 1 M KNO_3 onto the filter paper until it is saturated, but not so much that it dilutes the metal solutions.
5. Plug in the voltage probe and open Logger Pro; for the moment, do not attach it to anything. Check that the alligator clips are clean and dry. You should see a non-zero voltage displayed in the upper left corner. Touch the leads together and the voltage should go to close to zero. If not, see your instructor or TA. Adjust Data Collection

with an appropriate sampling rate (10 per min) to measure voltage changes with time collection over 10 minutes.

6. The positive lead (red) should be attached to the metal that has the most positive reduction potential. Connect the negative lead (black) to the metal with the least positive reduction potential. Explain why the voltage probe must be set up like this. Write a balanced reaction for the cell in your notebook.
7. At standard conditions, the voltage in this cell should be about 1.100 V. If we reduce the concentration of Cu^{2+}, the reaction should become less spontaneous and the voltage should drop. Adding an anion that precipitates the copper ion can reduce the copper ion concentration. Hydroxide ion will form insoluble (slightly soluble, actually) salts with many transition metal cations.
8. Select COLLECT to start the data collection on the cell and collect enough data points to get a good reading of the cell potential. Obtain the 6 M sodium hydroxide ($NaOH$).
9. Add a 2 mL portion of sodium hydroxide solution and using a stirring rod carefully stir the mixture to thoroughly mix the sodium hydroxide with the original copper solution. Collect sufficient data points to get a good reading of the new cell potential.
10. Repeat with two additional 2 mL portions of the sodium hydroxide solution. What can you say about the voltage as the copper ion concentration decreases? What does the voltage say about the spontaneity of the reaction?
11. Repeat the last two steps with the zinc solution, answering the questions as well.
12. Write the shorthand notation for this cell including symbols for phase changes, salt bridge, etc. Identify the anode and cathode.

Table 4. Half-reactions for zinc and copper reacting to generate overall cell-reaction and calculate E°.

Half-reaction	$\boldsymbol{E^{\circ}_{cell}}$ **(V)**
Cell-reaction	**Total** $\boldsymbol{E^{\circ}_{cell}}$ **(V)**

13. Remove the voltage probe from the metal electrodes. Wipe off any solution with a paper towel. Make sure the alligator clips are clean and dry.
14. Put the filter paper in the trash and the solutions in the waste container. Wash both beakers with soap and water.

WASTE AND CLEANUP

1. Put all waste in the solid or aqueous waste containers in the hood and clean all glassware with soap and an appropriate brush or scrubby.
2. Return your cleaned glassware and other items to the cart.

CONCLUSION

Be sure to summarize your findings from Parts A-D of today's experiment.

APPENDIX – Reduction Half Reactions

Selected Standard Reduction Potentials at 25 °C

Half-Reaction	***E*° (V)**
$F_{2(g)} + 2e^- \longrightarrow 2\ F^-_{(aq)}$	+2.866
$PbO_{2(s)} + SO_4^{2-}{}_{(aq)} + 4\ H^+_{(aq)} + 2e^- \longrightarrow PbSO_{4(s)} + 2\ H_2O_{(l)}$	+1.69
$MnO_4^-{}_{(aq)} + 8\ H^+_{(aq)} + 5e^- \longrightarrow Mn^{2+}_{(aq)} + 4\ H_2O_{(l)}$	+1.507
$Au^{3+}_{(aq)} + 3e^- \longrightarrow Au_{(s)}$	+1.498
$Cl_{2(g)} + 2e^- \longrightarrow 2\ Cl^-_{(aq)}$	+1.35827
$O_{2(g)} + 4\ H^+_{(aq)} + 4e^- \longrightarrow 2\ H_2O_{(l)}$	+1.229
$Pt^{2+}_{(aq)} + 2e^- \longrightarrow Pt_{(s)}$	+1.20
$Br_{2(aq)} + 2e^- \longrightarrow 2\ Br^-_{(aq)}$	+1.0873
$Ag^+_{(aq)} + e^- \longrightarrow Ag_{(s)}$	+0.7996
$Hg_2^{2+}{}_{(aq)} + 2e^- \longrightarrow 2\ Hg_{(l)}$	+0.7973
$Fe^{3+}_{(aq)} + e^- \longrightarrow Fe^{2+}_{(aq)}$	+0.771
$MnO_4^-{}_{(aq)} + 2\ H_2O_{(l)} + 3e^- \longrightarrow MnO_{2(s)} + 4\ OH^-_{(aq)}$	+0.558
$I_{2(s)} + 2e^- \longrightarrow 2\ I^-_{(aq)}$	+0.5355
$NiO_{2(s)} + 2\ H_2O_{(l)} + 2e^- \longrightarrow Ni(OH)_{2(s)} + 2\ OH^-_{(aq)}$	+0.49
$Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$	+0.34
$Hg_2Cl_{2(s)} + 2e^- \longrightarrow 2\ Hg_{(l)} + 2\ Cl^-_{(aq)}$	+0.26808
$AgCl_{(s)} + e^- \longrightarrow Ag_{(s)} + Cl^-_{(aq)}$	+0.22233
$Sn^{4+}_{(aq)} + 2e^- \longrightarrow Sn^{2+}_{(aq)}$	+0.151
$2\ H^+_{(aq)} + 2e^- \longrightarrow H_{2(g)}$	0.00
$Pb^{2+}_{(aq)} + 2e^- \longrightarrow Pb(s)$	−0.1262

Half-Reaction	$E°$ (V)
$Sn^{2+}_{(aq)} + 2e^- \longrightarrow Sn_{(s)}$	−0.1375
$Ni^{2+}_{(aq)} + 2e^- \longrightarrow Ni_{(s)}$	−0.257
$Co^{2+}_{(aq)} + 2e^- \longrightarrow Co_{(s)}$	−0.28
$PbSO_{4(s)} + 2e^- \longrightarrow Pb_{(s)} + SO_4^{2-}{}_{(aq)}$	−0.3505
$Cd^{2+}_{(aq)} + 2e^- \longrightarrow Cd_{(s)}$	−0.4030
$Fe^{2+}_{(aq)} + 2e^- \longrightarrow Fe_{(s)}$	−0.447
$Cr^{3+}_{(aq)} + 3e^- \longrightarrow Cr_{(s)}$	−0.744
$Mn^{2+}_{(aq)} + 2e^- \longrightarrow Mn_{(s)}$	−1.185
$Zn(OH)_{2(s)} + 2e^- \longrightarrow Zn_{(s)} + 2\ OH^-_{(aq)}$	−1.245
$Zn^{2+}_{(aq)} + 2e^- \longrightarrow Zn_{(s)}$	−0.7618
$Al^{3+}_{(aq)} + 3e^- \longrightarrow Al_{(s)}$	−1.662
$Mg^{2+}_{(aq)} + 2e^- \longrightarrow Mg_{(s)}$	−2.372
$Na^+_{(aq)} + e^- \longrightarrow Na_{(s)}$	−2.71
$Ca^{2+}_{(aq)} + 2e^- \longrightarrow Ca_{(s)}$	−2.868
$Ba^{2+}_{(aq)} + 2e^- \longrightarrow Ba_{(s)}$	−2.912
$K^+_{(aq)} + e^- \longrightarrow K_{(s)}$	−2.931
$Li^+_{(aq)} + e^- \longrightarrow Li_{(s)}$	−3.04

Table 16.1